

PRABAL KANTI KHAN

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PROFESSIONAL SUMMARY

B.Tech ECE graduate (2026) targeting the Graduate Engineer Technician (Assembly & Test) role at Micron. Hands-on with electrical and mechanical troubleshooting of non-functioning electro-mechanical equipment, equipment assembly and rebuilds, and diagnostic-tool checkouts. Comfortable interpreting blueprints, schematics, layout plans, and repair manuals, with module-level testing across 20+ lab deployments. Ready to relocate to Sanand, Gujarat and work rotational shifts.

TECHNICAL SKILLS

Troubleshooting & Diagnostics: Electrical/mechanical fault isolation on non-functioning equipment, component-level debugging, test and diagnostic equipment (multimeter, oscilloscope), equipment checkouts, computer hardware diagnostics, OS troubleshooting (Windows/Linux)

Assembly & Repair: Dismantling, adjusting, repairing, rebuilding, and wiring electro-mechanical systems to specification; relay panels, SMPS, sensor/actuator modules, Li-ion battery packs

Drawings & Documentation: Reading and interpreting blueprints, schematics, PCB/circuit layouts, rough sketches, and repair manuals; commissioning checklists and technical documentation

Microcontrollers & Programming: STM32 (STM32F446RE bare-metal C), ESP32, Arduino, Raspberry Pi; Embedded C/C++, C, Python

Protocols & Interfaces: I2C, SPI, UART, GPIO, LoRa (RYLR998), Wi-Fi, Bluetooth

EDUCATION & CERTIFICATIONS

B.Tech – Electronics & Communication Engineering, Heritage Institute of Technology June 2026 | DGPA: 7.98

Diploma – Electronics & Instrumentation Engineering, Siliguri Govt. Polytechnic July 2022 | GPA: 7.7

Certification: Microcontroller Embedded C Programming — Udemy (FastBitLab, Kiran Nayak) June 2026

PROFESSIONAL EXPERIENCE

STEM Innovation Engineer | STEMROBO Technologies Pvt. Ltd. Feb 2022 – Jul 2023

- Assembled, wired, and functionally tested embedded hardware across 20+ innovation labs — MCU boards, relay panels, sensor modules, and actuator circuits — constructed to layout plans before commissioning.
- Performed electrical and mechanical troubleshooting with component-level fault isolation on non-functioning equipment using multimeter and diagnostic instruments; dismantled and rebuilt faulty setups.
- Standardised build/repair checklists and test documentation for repeatable quality; trained 500+ users with 90% workshop satisfaction.

IoT & Django Intern | Euphoria GenX (via Heritage Institute of Technology) Jun – Jul 2025

- Developed real-time IoT dashboards linking live sensor data to a Django backend; validated end-to-end signal flow from physical input to visualisation.
- Traced and resolved faults across the data-acquisition and automation chain.

PROJECTS

Agriculture Automation System (LoRa) | Team Leader | Final Year Project | YouTube - @TechnicZ 2025 – 2026

- Designed and assembled a dual-node system: Farm Node (ESP32 + RYLR998) reads DHT22, MQ135, and soil-moisture sensors and drives Peltier AC, humidifier, exhaust fan, and irrigation pump via two 4-channel relays.
- Built custom BC547 transistor control circuits and the full power architecture (12V/20A SMPS, dual 18650 Li-ion + 6V solar/TP4056) from technical drawings for 24/7 off-grid operation.
- Ran diagnostic checkouts and module-level pass/fail validation before full integration; added an MQTT cloud dashboard over a Wi-Fi gateway.

DIY Bluetooth Speaker | Personal Project | YouTube - @TechnicZ

- Fabricated a custom speaker enclosure from PVC piping, handling mechanical fitting and assembly to a planned layout.
- Connected a 4S Li-ion battery pack to the audio modules, confirming voltage and polarity by multimeter before power-up.

Smart Home Automation System (Arduino + Bluetooth) | Team Leader | YouTube - @TechnicZ May – Jun 2022

- Mounted and wired a multi-module relay control panel — Bluetooth switching, IR doorbell, LDR lighting, automatic tank cutoff (BC547 + relay), and solar power.
- Verified each module independently before integrating the full working unit; authored complete Embedded C firmware.

ADDITIONAL INFORMATION

Languages: Bengali (Native), English & Hindi (Fluent) | Batch: 2026 | No Active Backlogs | Relocation: Ready for Sanand, Gujarat | Shifts: Willing (rotational)